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**Tutorial No.: 7**  
**Engineering Mechanics (ME-101)**  
**Date: 11/03/2024 and Time: 7.55 to 8.50 AM**

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**Question 1:** The motion of a particle is defined by the position vector  $\mathbf{r} = A(\cos t + t \sin t)\mathbf{i} + A(\sin t - t \cos t)\mathbf{j}$ , as shown in **Figure 1**, where  $t$  is expressed in seconds. Determine the values of  $t$  for which the position vector and the acceleration are (a) perpendicular; (b) parallel.

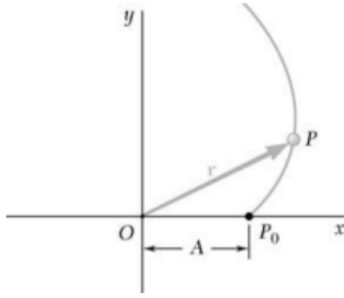


Figure 1

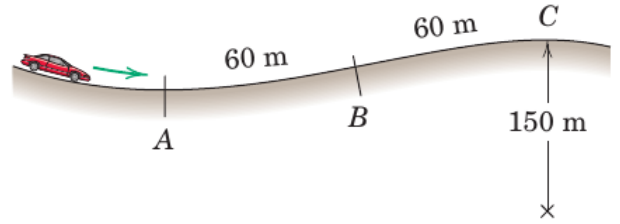


Figure 2

**Question 2:** To anticipate the dip and hump in the road (refer **Figure 2**), the driver of a car applies her brakes to produce a uniform deceleration. Her speed is 100 km/h at the bottom  $A$  of the dip and 50 km/h at the top  $C$  of the hump, which is 120 m along the road from  $A$ . If the passengers experience a total acceleration of  $3 \text{ m/s}^2$  at  $A$  and if the radius of curvature of the hump at  $C$  is 150 m, calculate (a) the radius of curvature at  $A$ , (b) the acceleration at the inflection point  $B$ , and (c) the total acceleration at  $C$ .

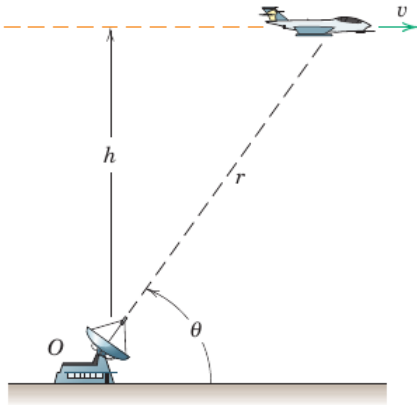


Figure 3

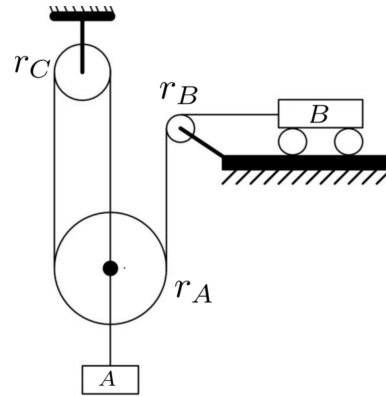


Figure 4

**Question 3:** A jet plane flying at a constant speed  $v$  at an altitude  $h = 10 \text{ km}$  is being tracked by radar located at  $O$  directly below the line of flight, shown in **Figure 3**. If the angle  $\theta$  is decreasing at the rate of  $0.020 \text{ rad/s}$  when  $\theta = 60^\circ$ , determine the value of  $\ddot{r}$  at this instant and the magnitude of the velocity  $v$  of the plane.

**Question 4:** If the block  $B$  has a leftward velocity of 1.2 m/s, determine the velocity of block  $A$ , as shown in **Figure 4**.

**Question 5:** The paint spraying robot (depicted in **Figure 5**) is programmed to paint a production line of curved surface  $A$  (seen on edge). The length of the telescoping arm is controlled according to  $b = 0.3 \sin(\pi t/2)$  where  $b$  is in meter and  $t$  is in seconds. Simultaneously, the arm is programmed to rotate according to  $\theta = \frac{\pi}{4} + \frac{\pi}{8} \sin(\pi t/2)$  rad. Calculate the magnitude of the velocity and acceleration of the nozzle  $N$  at  $t = 1$  sec.

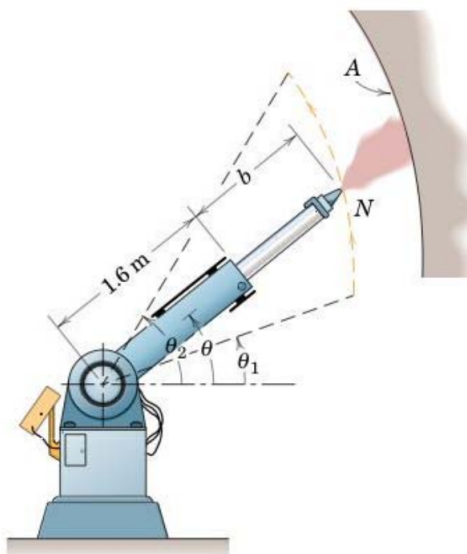


Figure 5

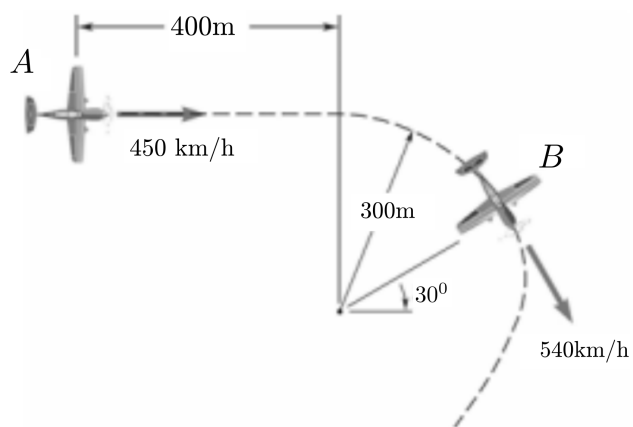


Figure 6

**Question 6:** At a given instant in an airplane race (refer **Figure 6**), airplane  $A$  is flying horizontally in a straight line, and its speed is being increased at the rate of  $8 \text{ m/s}^2$ . Airplane  $B$  is flying at the same altitude as airplane  $A$  and, as it rounds a pylon, is following a circular path of 300 m radius. Knowing that the given instant the speed of  $B$  is being decreased at the rate of  $3 \text{ m/s}^2$ , determine, for the positions shown, (a) the velocity of  $B$  relative to  $A$ , (b) the acceleration of  $B$  relative to  $A$ .